

DBMS: Unit 1 Relational Model and SQL Calculus

VTU 2022 Scheme CBCS - BCS501 Module 1

--- MODULE OVERVIEW & LEARNING OBJECTIVES ---

1. Relational Database Concepts: Relations, Tuples, Attributes, and Domains
2. Relational Integrity Constraints: Domain, Entity Integrity, and Referential Integrity
3. Relational Algebra Operations: Select, Project, Cartesian Product, Join, and Division
4. SQL DDL, DML, DCL, and TCL commands with complex correlated subqueries
5. ER-to-Relational Mapping: Converting Entities, Weak Entities, and M:N relationships

--- IMPORTANT VTU 10-MARK QUESTIONS ---

- Q1. Write Relational Algebra queries for the Given Company schema [10 Marks]
- Q2. Explain 1NF, 2NF, 3NF, and BCNF normalization with proofs [10 Marks]